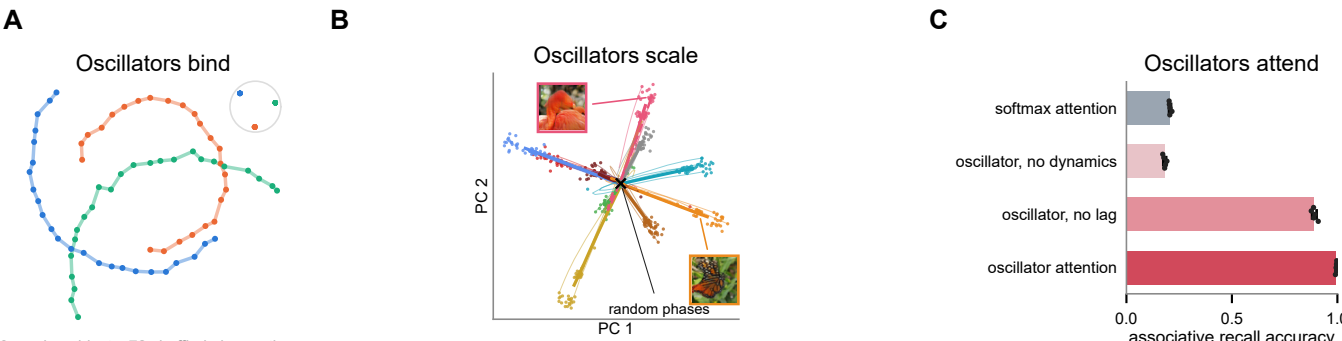

A
Oscillators bind
3 moving objects, 72 shuffled observations;
colour = phase group (100% correct)
B
Oscillators scale
PC 2
PC 1
random phases
Un-0, 16,384 oscillators: random phases → class clusters
(10 ImageNet classes; class decoding 10% → 97%)
C
Oscillators attend
softmax attention
oscillator, no dynamics
oscillator, no lag
oscillator attention
0.0
0.5
1.0
associative recall accuracy
1 layer each, ~58k parameters; chance 3%